

**UNIVERSITY OF SCIENCE AND TECHNOLOGY OF HANOI**  
**DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY**



# **BACHELOR THESIS**

By

**NGUYEN DUY NGOC**

Title:

**Development of an Internal AI Chatbot  
for Company Knowledge Retrieval**

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To whom it may concern,

I, **Trần Đình Chí**, hereby certify that the Bachelor Thesis entitled “**Development of an Internal AI Chatbot for Company Knowledge Retrieval**” submitted by Mr. **Nguyen Duy Ngoc** has been completed under my supervision.

I confirm that this thesis meets the requirements for presentation and evaluation before the Internship/Thesis Evaluation Committee in the academic year 2025–2026.

Hanoi, \_\_/\_\_/\_\_\_\_

**Supervisor’s signature**

# Contents

|                                                        |             |
|--------------------------------------------------------|-------------|
| <b>List of Abbreviations</b>                           | <b>iv</b>   |
| <b>Acknowledgements</b>                                | <b>vii</b>  |
| <b>Abstract</b>                                        | <b>viii</b> |
| <b>1 Introduction</b>                                  | <b>1</b>    |
| 1.1 Context and Problem . . . . .                      | 1           |
| 1.2 Related Work . . . . .                             | 1           |
| 1.3 Contributions . . . . .                            | 2           |
| 1.4 Research Questions and Scope . . . . .             | 2           |
| <b>2 Objectives</b>                                    | <b>4</b>    |
| <b>3 Materials and Methods</b>                         | <b>5</b>    |
| 3.1 System Architecture . . . . .                      | 5           |
| 3.2 Knowledge Base and Indexing . . . . .              | 6           |
| 3.3 Hybrid Retrieval . . . . .                         | 7           |
| 3.4 Answer Generation . . . . .                        | 8           |
| 3.5 Implementation and Privacy Controls . . . . .      | 8           |
| 3.6 Synthetic Data Generation and Validation . . . . . | 9           |
| 3.7 QLoRA Training and Direct Evaluation . . . . .     | 10          |
| 3.8 Evaluation Protocol . . . . .                      | 11          |
| <b>4 Results and Discussion</b>                        | <b>13</b>   |
| 4.1 Retrieval Results . . . . .                        | 13          |
| 4.2 Fine-Tuning Results . . . . .                      | 13          |
| 4.3 Role of RAG After Fine-Tuning . . . . .            | 14          |
| 4.4 Limitations and Threats to Validity . . . . .      | 15          |

|                                     |           |
|-------------------------------------|-----------|
| <b>5 Conclusion and Perspective</b> | <b>16</b> |
| <b>A Reproducibility Map</b>        | <b>17</b> |

## List of Abbreviations

| Abbreviation | Full Form                                     |
|--------------|-----------------------------------------------|
| AI           | Artificial Intelligence                       |
| BM25         | Best Matching 25                              |
| CPU          | Central Processing Unit                       |
| GGUF         | GPT-Generated Unified Format                  |
| GPU          | Graphics Processing Unit                      |
| HR           | Human Resources                               |
| LLM          | Large Language Model                          |
| LoRA         | Low-Rank Adaptation                           |
| NLP          | Natural Language Processing                   |
| QLoRA        | Quantized Low-Rank Adaptation                 |
| RAG          | Retrieval-Augmented Generation                |
| RRF          | Reciprocal Rank Fusion                        |
| UI           | User Interface                                |
| USTH         | University of Science and Technology of Hanoi |
| VRAM         | Video Random-Access Memory                    |

## Lists of Tables and Figures

## List of Tables

|     |                                                                                     |    |
|-----|-------------------------------------------------------------------------------------|----|
| 3.1 | Principal implementation components. . . . .                                        | 6  |
| 3.2 | Vietnamese HR knowledge base: source documents and generated data coverage. . . . . | 7  |
| 3.3 | Validation summary for the H200-generated dataset. . . . .                          | 10 |
| 3.4 | Accepted Q&A pairs by source topic. . . . .                                         | 10 |
| 3.5 | H200 QLoRA training configuration and outcome. . . . .                              | 11 |
| 4.1 | Retrieval comparison on ten labelled questions. . . . .                             | 13 |
| 4.2 | Direct, retrieval-free evaluation on 20 fixed questions. . . . .                    | 14 |
| 4.3 | Representative fine-tuned outputs requiring factual review. . . . .                 | 14 |
| A.1 | Local evidence artifacts used in this report. . . . .                               | 17 |

**List of Figures**

3.1    Offline Vietnamese HR RAG architecture. . . . . 5

3.2    Hybrid BM25–vector retrieval workflow using reciprocal rank fusion. . . . . 8

4.1    Compact summary of the ten-query internal retrieval test. . . . . 13

4.2    Local Vietnamese HR chatbot statistics after rebuilding the index. . . . . 15

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## Abstract

This thesis presents a fully local Retrieval-Augmented Generation (RAG) chatbot designed to answer Vietnamese Human Resources (HR) policy-related questions. A total of 20 HR policy documents were processed and indexed using multilingual sentence embeddings and ChromaDB. The proposed system applies a hybrid retrieval approach that combines dense similarity search and BM25 ranking using Reciprocal Rank Fusion (RRF). The retrieved relevant passages are then provided to a Phi-3-Mini GGUF language model to generate responses. In a small retrieval evaluation with 10 manually labelled queries, the hybrid retrieval method achieved 100% top-1 accuracy, while the dense retrieval baseline achieved 60%.

This study also explores domain adaptation through model fine-tuning. The Qwen2.5-72B-Instruct model was used to generate 4,900 unique question-answer pairs covering 20 HR policy topics. A keyword-based validation process showed that 82.4% of the generated questions were clearly related to HR topics, with no exact duplicate questions or predefined irrelevant cases detected. QLoRA fine-tuning was performed on an H200 GPU and completed three training epochs in 5.3 minutes. For 20 direct questions without retrieval support, the fine-tuned model improved the automated format and relevance score from 0.62 to 0.70, while reducing the average response time from 3.9 seconds to 0.9 seconds.

However, manual evaluation still identified incorrect policy information and inconsistent answers. Therefore, the automated evaluation score cannot be considered sufficient evidence of factual correctness. The final system keeps RAG as the main approach for reliable knowledge retrieval, while fine-tuning is used as an additional improvement method. The experimental results demonstrate that hybrid retrieval provides better matching performance for short queries, and that data quality and evaluation methods are important factors in building reliable local AI assistants.

**Keywords:** Retrieval-Augmented Generation, Vietnamese NLP, BM25, Reciprocal Rank Fusion, QLoRA, Local Language Model.

# 1. Introduction

## 1.1 Context and Problem

Company policies are frequently distributed as long PDF handbooks. Employees must locate rules on leave, salary, insurance, remote work, or disciplinary procedures, while HR staff repeatedly answer similar questions. Keyword search helps only when users know the exact wording used by the document. The problem is harder in Vietnamese because short colloquial questions may omit formal policy terms.

Cloud language models offer fluent interaction but can expose confidential documents, require network access, and introduce recurring cost. A local system avoids these constraints, yet a small model cannot be expected to memorize each company’s policy or reliably infer facts from its parameters. RAG addresses this limitation by retrieving relevant passages at query time and conditioning generation on those passages [1].

This work asks whether a compact, offline pipeline can retrieve Vietnamese HR policy passages accurately enough to support a local responder. It also asks whether QLoRA domain adaptation improves a small model when trained on a quality-gated synthetic dataset. The second question is deliberately separated from deployment: an ungrounded model may produce plausible but incorrect policy values even when its language and speed improve.

## 1.2 Related Work

RAG combines non-parametric retrieval with neural generation, allowing evidence to be updated without retraining [1]. Dense retrievers encode queries and passages into a vector space. Sentence-BERT made this practical for semantic similarity [5], while multilingual encoders extend the approach across languages.

BM25 remains a strong lexical ranking method [2]. Dense retrieval captures meaning but may miss exact terms in short queries; BM25 has the opposite strengths. Reciprocal Rank Fusion (RRF) combines ranked lists without requiring calibrated scores [3]. This makes it useful when vector similarity and BM25 scores have different scales.

Phi-3-Mini is a compact instruction-tuned model intended for constrained local deployment [4]. LoRA adds trainable low-rank matrices while freezing base weights [7]; QLoRA applies this method to a quantized base model to reduce training memory [6]. These techniques make adap-

tation affordable, but they do not guarantee that synthetic labels are correct or that automatic relevance checks measure factual accuracy.

Vietnamese introduces additional retrieval difficulty because whitespace does not always correspond to word boundaries. PhoBERT illustrates the value of language-aware pretraining for Vietnamese NLP [10], while ViQuAD provides a reference point for Vietnamese machine-reading evaluation [11]. The present implementation uses a multilingual encoder and simple whitespace BM25 tokenization, which is efficient but remains a limitation.

### 1.3 Contributions

The main contributions of this thesis are summarized as follows:

1. **Private and Offline Vietnamese HR RAG Pipeline:** This thesis develops a private, offline RAG pipeline for Vietnamese HR knowledge retrieval, integrating local embeddings, ChromaDB, hybrid retrieval, and Phi-3-Mini GGUF inference.
2. **Dense and Hybrid Retrieval Comparison:** This thesis compares dense and hybrid retrieval methods for short Vietnamese HR queries in an internal chatbot scenario.
3. **Quality-Gated Vietnamese HR Q&A Dataset:** This thesis constructs a quality-gated dataset of 4,900 unique Vietnamese HR question-answer pairs for chatbot evaluation and experimentation.
4. **H200 QLoRA Fine-Tuning Experiment:** This thesis conducts a measured QLoRA experiment on the H200 platform using Python and JSON-based evaluation workflows, while distinguishing automated relevance indicators from factual policy correctness.

### 1.4 Research Questions and Scope

The evaluation is structured around three research questions. **RQ1** examines whether the combination of lexical and dense rankings improves top-document retrieval for short Vietnamese HR questions. **RQ2** investigates whether a controlled synthetic dataset can shift Phi-3-Mini toward Vietnamese HR-specific language while reducing direct inference latency. **RQ3** assesses whether automated evaluation indicators are sufficient to determine factual policy correctness.

RQ1 is addressed through a paired comparison using the same ten labeled queries. RQ2 uses identical 20-question prompts for both the base and adapted models. RQ3 requires manual inspection of the generated outputs and analysis of the scoring code. The scope of this thesis does not include legal assessment, deployment in a real company, or a formal human-subject

study. This boundary is important because a technically functional chatbot is not equivalent to a validated HR decision-support system.

## 2. Objectives

The project has four objectives:

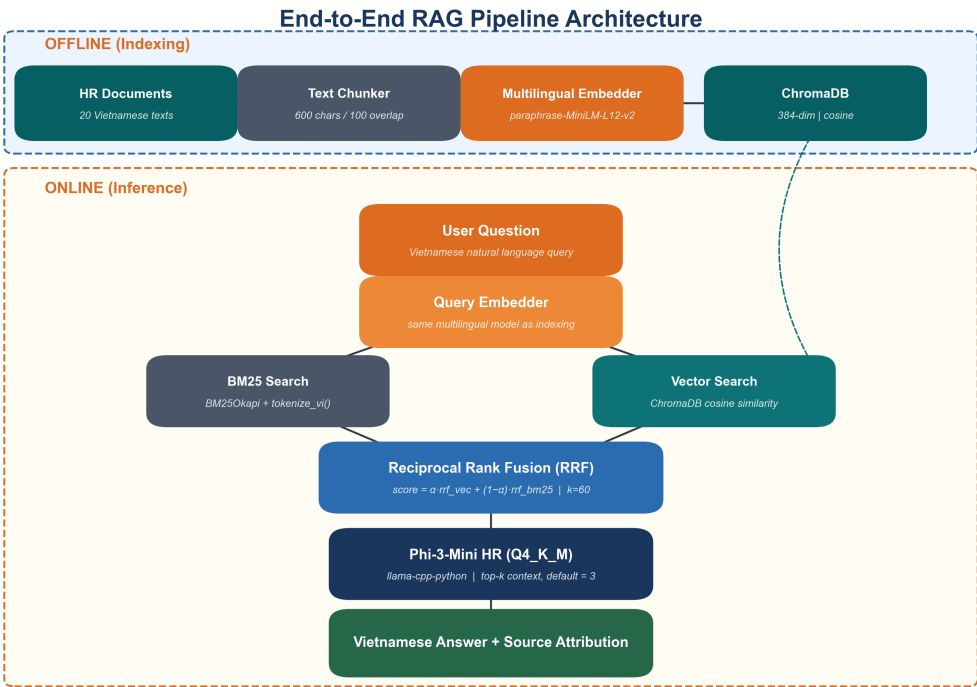
1. Design and implement a fully local Vietnamese HR question-answering system in which source documents remain on the user’s machine;
2. Compare dense retrieval with BM25–vector fusion for short HR-related queries;
3. Construct and validate a synthetic domain-specific dataset, then assess the QLoRA adaptation of Phi-3-Mini;
4. Evaluate the system transparently, including incorrect outputs and the limitations of small test sets and heuristic metrics.

The implemented architecture is considered successful when it retrieves a relevant policy source, constructs a grounded prompt, and returns a Vietnamese answer with a source reference. The fine-tuning experiment is evaluated separately because its direct, retrieval-free test measures model adaptation rather than end-to-end RAG correctness.

### 3. Materials and Methods

#### 3.1 System Architecture

Figure 3.1 illustrates the offline processing and query paths of the proposed system. During indexing, Vietnamese HR policy documents are split into overlapping chunks, embedded, and stored in ChromaDB. During querying, vector retrieval and BM25 retrieval are performed over the same document corpus. Reciprocal rank fusion (RRF) is then applied to select the top-ranked passages, which are inserted into a Phi-3 chat prompt. The model is executed through llama-cpp-python, ensuring that no company document is sent to an external inference API.



**Figure 3.1:** *Offline Vietnamese HR RAG architecture.*

**Table 3.1:** *Principal implementation components.*

| Stage          | Technology                                                 | Role                                 |
|----------------|------------------------------------------------------------|--------------------------------------|
| Documents      | 20 UTF-8 policy files                                      | Vietnamese HR knowledge source       |
| Chunking       | Paragraph-aware splitter, about 300 characters, 50 overlap | Preserves local policy units         |
| Embedding      | multilingual MiniLM, 384 dimensions                        | Dense semantic representation        |
| Vector store   | ChromaDB                                                   | Persistent cosine search             |
| Lexical search | BM25Okapi                                                  | Exact-term ranking                   |
| Fusion         | RRF, $k = 60$                                              | Combines vector and lexical ranks    |
| Generation     | Phi-3-Mini HR Q4_K_M                                       | Local Vietnamese response generation |

## 3.2 Knowledge Base and Indexing

The knowledge base covers 20 topics, including leave, salary, contracts, social insurance, maternity, overtime, remote work, IT security, workplace safety, and anti-harassment. The Vietnamese Labor Code provides general legal context for these topics [9]; however, the demonstration policies are not legal interpretations. Eight initial documents were manually authored, while twelve additional documents were generated to broaden topic coverage. The corpus represents a demonstration company rather than the policies of a real employer; therefore, the generated answers must not be interpreted as legal advice.

After cleaning, the documents produced 258 text chunks. Each chunk retains its source name for attribution. The multilingual MiniLM encoder maps text into 384-dimensional vectors, which are stored in a persistent ChromaDB collection [8]. Table 3.2 summarizes the corpus coverage and the final number of accepted Q&A pairs produced from each source.

**Table 3.2:** Vietnamese HR knowledge base: source documents and generated data coverage.

| No.          | Policy topic                       | Source document               | Origin       | Q&A pairs    |
|--------------|------------------------------------|-------------------------------|--------------|--------------|
| 1            | Leave policy                       | chinh_sach_nghi_phep.txt      | Manual draft | 264          |
| 2            | Employment contracts               | hop_dong_lao_dong.txt         | Manual draft | 262          |
| 3            | Salary policy                      | chinh_sach_luong.txt          | Manual draft | 184          |
| 4            | Company regulations                | noi_quy_cong_ty.txt           | Manual draft | 338          |
| 5            | Labour discipline                  | ky_luat_lao_dong.txt          | Manual draft | 256          |
| 6            | Employee benefits                  | phuc_loi_nhan_vien.txt        | Manual draft | 282          |
| 7            | Recruitment and training           | tuyen_dung_dao_tao.txt        | Manual draft | 286          |
| 8            | Social insurance                   | bao_hiem_xa_hoi.txt           | Manual draft | 122          |
| 9            | Overtime policy                    | lam_them_gio.txt              | Minimax-M2.7 | 240          |
| 10           | Maternity and women employees      | nghi_thai_san_va_phu_nu.txt   | Minimax-M2.7 | 242          |
| 11           | Remote work                        | chinh_sach_lam_viec_tu_xa.txt | Minimax-M2.7 | 327          |
| 12           | Business travel                    | cong_tac_va_di_chuyen.txt     | Minimax-M2.7 | 160          |
| 13           | Performance review                 | danh_gia_hieu_suat.txt        | Minimax-M2.7 | 254          |
| 14           | IT security                        | bao_mat_thong_tin_it.txt      | Minimax-M2.7 | 230          |
| 15           | Special holidays and leave         | nghi_le_phep_dac_biet.txt     | Minimax-M2.7 | 287          |
| 16           | Anti-harassment and discrimination | quay_roi_phan_biet_doi_xu.txt | Minimax-M2.7 | 300          |
| 17           | Resignation and handover           | nghi_viec_va_ban_giao.txt     | Minimax-M2.7 | 287          |
| 18           | Anti-corruption                    | phong_chong_tham_nhung.txt    | Minimax-M2.7 | 179          |
| 19           | Workplace safety                   | suc_khoe_an_toan_lao_dong.txt | Minimax-M2.7 | 219          |
| 20           | Allowances and subsidies           | phu_cap_va_tro_cap.txt        | Minimax-M2.7 | 181          |
| <b>Total</b> |                                    |                               |              | <b>4,900</b> |

### 3.3 Hybrid Retrieval

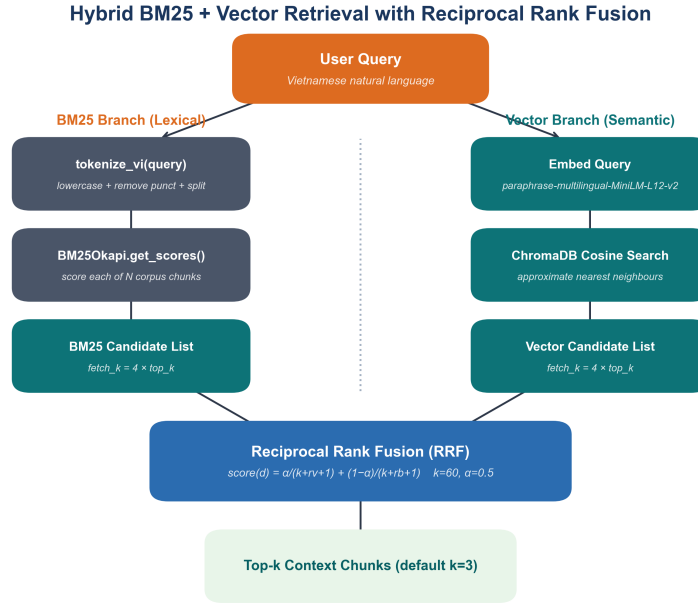
Dense search retrieves semantically related text, while BM25 emphasizes exact query-term matching. To combine the advantages of both methods, their ranked lists are fused using reciprocal rank fusion (RRF), defined as follows:

$$S(d) = \alpha \frac{1}{k + r_v(d) + 1} + (1 - \alpha) \frac{1}{k + r_b(d) + 1}, \quad (3.1)$$

where  $r_v$  and  $r_b$  are the vector and BM25 ranks,  $k = 60$ , and  $\alpha = 0.5$ . The implementation returns a configurable top- $k$  context; the deployed answer path defaults to three chunks. Because



RRF uses ranks rather than raw scores, no cross-system score calibration is needed.



**Figure 3.2:** Hybrid BM25–vector retrieval workflow using reciprocal rank fusion.

### 3.4 Answer Generation

The responder uses Phi-3-Mini-4k-Instruct in GGUF Q4\_K\_M format [4]. The prompt includes a Vietnamese instruction, retrieved chunks with source labels, and the user question formatted with Phi-3 chat tokens. Generation is configured with a low temperature and stops at the model’s chat boundary. Source grounding is achieved through the retrieved context and prompt structure, rather than by assuming that the model parameters contain correct company-specific rules.

### 3.5 Implementation and Privacy Controls

The implementation separates ingestion, retrieval, and generation into Python modules. PDF extraction and chunking produce serializable records; the embedding module manages the ChromaDB collection; the hybrid retriever uses the same chunk set for both BM25 and vector ranking; and the responder wraps `llama-cpp-python`. This modular design allows retrieval tests to be performed without loading the 2.32 GB generator model and makes failure sources easier to identify.

At startup, the application checks whether the GGUF model file and vector collection are avail-

able. A user question is normalized, embedded, and sent to both retrieval branches. The fused top-ranked passages, defaulting to three context chunks in the answer path, are formatted with their source names before local generation. The interface returns the generated text together with source metadata, allowing the user to inspect the underlying policy rather than treating the model output as an authoritative decision.

Privacy follows from the local data flow: policy text, embeddings, prompts, and answers remain on the machine during normal operation. External computers were used only to create the synthetic research dataset and train the adapter; it was not part of the deployed inference path. A real organization would additionally require access control, encrypted storage, audit logging, retention policies, and review procedures for uploaded documents. These controls are outside the prototype scope but are necessary before production use.

The principal failure control is prompt-instructed abstention when the retrieved context is insufficient. However, this mechanism is weaker than formal entailment verification. Source display therefore remains important, especially for numerical benefits or legal questions. No model response should replace an approved policy document or professional HR judgment.

## **3.6 Synthetic Data Generation and Validation**

An initial fine-tuning attempt used a web-crawled dataset that mixed unrelated material with HR content. Its outputs included references to Medicare and other off-domain subjects. Rather than treating that run as representative of QLoRA, the experiment was repeated with data generated only from the 20 controlled HR documents.

Qwen2.5-72B-Instruct-AWQ ran with vLLM on a Vast.ai H200. Prompts requested natural Vietnamese questions and source-grounded answers for each document. Generation resumed by document and batch, normalized questions were deduplicated, and the accepted output stopped at the available final total of 4,900 pairs.

Validation checked JSON structure, length, exact normalized question duplicates, predefined off-domain red flags, and a Vietnamese HR keyword set. These checks are reproducible but shallow: a pair labelled “unclear” may still be relevant, and a pair containing an HR keyword may still contain a wrong policy value.

**Table 3.3:** *Validation summary for the H200-generated dataset.*

| Measure                           | Result        |
|-----------------------------------|---------------|
| Accepted Q&A pairs                | 4,900         |
| Source documents                  | 20            |
| Keyword-classified HR-relevant    | 4,037 (82.4%) |
| Unclear under keyword rules       | 863           |
| Predefined red flags              | 0             |
| Exact duplicate questions         | 0             |
| Short questions (< 10 characters) | 4             |
| Short answers (< 10 characters)   | 65            |

All 20 source topics remained represented after deduplication. The distribution is not uniform because document length and parseable model output differed by batch. Table 3.4 records the final counts and makes this imbalance explicit rather than treating 4,900 as a uniformly stratified sample.

**Table 3.4:** *Accepted Q&A pairs by source topic.*

| Topic                | Pairs | Topic                | Pairs        |
|----------------------|-------|----------------------|--------------|
| Social insurance     | 122   | IT security          | 230          |
| Remote work          | 327   | Salary policy        | 184          |
| Leave policy         | 264   | Business travel      | 160          |
| Performance review   | 254   | Employment contracts | 262          |
| Labour discipline    | 256   | Overtime             | 240          |
| Special leave        | 287   | Maternity            | 242          |
| Resignation/handover | 287   | Company regulations  | 338          |
| Anti-corruption      | 179   | Allowances           | 181          |
| Employee benefits    | 282   | Anti-harassment      | 300          |
| Workplace safety     | 219   | Recruitment/training | 286          |
|                      |       | <b>Total</b>         | <b>4,900</b> |

### 3.7 QLoRA Training and Direct Evaluation

The second run adapted unsloth/phi-3-mini-4k-instruct-bnb-4bit using QLoRA. Table 3.5 records the configuration from the training log. The LoRA adapter was saved, merged, and recovered as a 2.32 GB HR-labelled Q4\_K\_M GGUF. The artifact’s metadata identifies the fine-tune as hr.

**Table 3.5:** *H200 QLoRA training configuration and outcome.*

| Parameter                   | Value                       |
|-----------------------------|-----------------------------|
| Examples / epochs           | 4,900 / 3                   |
| Sequence length             | 2,048 tokens                |
| Device batch / accumulation | 8 / 2 (effective 16)        |
| LoRA rank / alpha           | 16 / 16                     |
| Learning rate / schedule    | $2 \times 10^{-4}$ / cosine |
| Quantized optimizer         | 8-bit AdamW                 |
| Training time               | 316 seconds (5.3 minutes)   |
| Final training loss         | 0.5093                      |
| Adapter size                | 119.6 MB                    |
| GGUF size                   | 2.32 GB                     |

Base and fine-tuned models answered the same 20 HR questions without retrieval. The evaluation assigned one point when an answer contained an HR keyword, used Vietnamese characters, was non-empty, and contained none of eight predefined off-domain/refusal strings; a partial score was possible. Consequently, this is a format and topicality indicator, not semantic fact checking. Greedy generation and per-question wall-clock latency were also recorded.

### 3.8 Evaluation Protocol

Retrieval and generation were evaluated separately to avoid attributing one component’s failure to another. The retrieval set paired ten questions with a gold source-document label. Accuracy@1 and Mean Reciprocal Rank were calculated as:

$$\text{Acc@1} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}[r_i = 1], \quad \text{MRR} = \frac{1}{N} \sum_{i=1}^N \frac{1}{r_i}, \quad (3.2)$$

where  $r_i$  is the rank of the labelled document. Dense and hybrid retrieval used the same corpus and queries.

For direct generation, each model received the same 20 questions and no policy context. Decoding was deterministic. The scoring function checked HR-keyword presence, Vietnamese-character frequency, minimum length, and eight literal off-domain/refusal flags. Latency measured one generation call on the H200 evaluation environment. Since answer correctness was not compared with a structured gold answer, manual error analysis is reported alongside the score.

The two evaluations answer different questions. Retrieval metrics measure whether evidence can be found; direct generation metrics measure topical form and speed after adaptation. Neither alone measures end-to-end user trust. A complete future protocol should combine retrieval, source-grounded generation, factual scoring, and human review on a larger held-out set.

All evaluation scripts are implemented in Python 3.10. The retrieval and generation tests use `pandas` for data aggregation, the built-in `re` module for keyword matching, and `time` for latency measurement. Input queries and outputs are stored in JSONL format to ensure reproducibility. The scoring logic is encapsulated in `scripts/validate_qa.py` and `src/hybrid_retriever.py`, as listed in Appendix A.

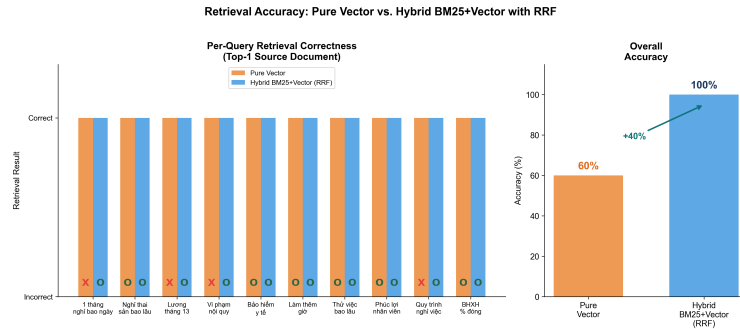
## 4. Results and Discussion

### 4.1 Retrieval Results

The internal retrieval test contains ten Vietnamese HR questions with manually assigned source documents. Dense search returned the correct top document for six questions, while hybrid retrieval returned it for all ten. MRR improved from 0.72 to 1.00 (Table 4.1). This is evidence for the tested queries, not a population-level accuracy estimate.

**Table 4.1:** *Retrieval comparison on ten labelled questions.*

| Metric                | Dense vector | Hybrid RRF   |
|-----------------------|--------------|--------------|
| Correct top-1 sources | 6/10         | <b>10/10</b> |
| Top-1 accuracy        | 60%          | <b>100%</b>  |
| Mean Reciprocal Rank  | 0.72         | <b>1.00</b>  |



**Figure 4.1:** *Compact summary of the ten-query internal retrieval test.*

The largest gains occurred for brief queries such as “one month has how many leave days?” For such wording, BM25 preserved exact leave-related terms while the dense model sometimes selected a salary or general-benefit document. RRF also retained dense evidence for longer paraphrases, so lexical matching did not replace semantic retrieval. The main uncertainty is sample size: ten questions cannot establish robustness across all employee language.

### 4.2 Fine-Tuning Results

Table 4.2 reports the direct model comparison. Fine-tuning raised the automated score by 0.07 in the report’s unrounded calculation, increased the HR-keyword flag from 13 to 14 answers,

and reduced mean latency by approximately 77%. Neither model triggered the narrow list of off-domain strings.

**Table 4.2:** *Direct, retrieval-free evaluation on 20 fixed questions.*

| Metric              | Base  | Fine-tuned | Change |
|---------------------|-------|------------|--------|
| Automated score     | 0.62  | 0.70       | +0.07  |
| HR-keyword relevant | 13/20 | 14/20      | +1     |
| Flag-list matches   | 0/20  | 0/20       | 0      |
| Mean latency        | 3.9 s | 0.9 s      | −3.0 s |

These numbers show that adaptation changed model behaviour, but qualitative inspection prevents a positive factual-accuracy conclusion. Examples include five days of annual leave, 0.05% social-insurance contribution, 500 VND lunch allowance, and a two-to-three-million-VND answer to a question about recruitment rounds. The flag-list metric recorded none of these as hallucinations because it only searched for words such as “Medicare,” “election,” and “Japan.”

**Table 4.3:** *Representative fine-tuned outputs requiring factual review.*

| Question topic     | Observed answer or issue                                        |
|--------------------|-----------------------------------------------------------------|
| Annual leave       | States a five-day annual maximum without source grounding.      |
| Maternity leave    | Answers “60 days per year,” confusing maternity and sick leave. |
| Social insurance   | Gives a 0.05% salary contribution.                              |
| Recruitment rounds | Answers with a monetary range rather than a number of rounds.   |
| Lunch allowance    | Gives 500 VND per working day, an implausible value.            |
| Performance review | Switches to an English list despite a Vietnamese question.      |

The experiment therefore has a mixed outcome. It is clearly better than the first contaminated-data run: responses are generally on the HR topic and are faster. It is not reliable enough to answer policy questions from parameters alone. Possible causes include synthetic-label errors, contradictory policy paraphrases, limited coverage of numeric facts, and a small model learning surface patterns rather than source-grounded reasoning.

### 4.3 Role of RAG After Fine-Tuning

The direct evaluation intentionally omitted retrieval, whereas the deployed system supplies policy passages. Fine-tuning and RAG answer different needs: fine-tuning can improve register, response structure, and domain vocabulary; RAG supplies current facts and source attribution.

The observed factual errors support keeping retrieval as the authority. A future comparison should run the base and fine-tuned GGUF models through the identical RAG pipeline and score answers against source-derived references.

Figure 4.2 shows the local CLI after rebuilding the controlled Vietnamese index. The statistics view exposes the model file, 258 indexed chunks, embedding dimension, and hybrid retrieval configuration. During normal question answering, source labels allow users to verify an answer; this is more defensible than presenting model confidence as policy correctness.

```

Vietnamese HR RAG Chatbot - Local Terminal

$ python cli_demo.py models/phi-3-mini.gguf
Vector store: 258 documents loaded
Model: phi-3-mini.gguf | Retrieval: hybrid BM25+vector RRF

You: Mức lương cơ bản là bao nhiêu
INFO: BM25 index ready: 258 chunks indexed.

Bot: Mức lương cơ bản bao gồm:
- Nhân viên mới / Thực tập sinh: 8.000.000 - 12.000.000 VND/tháng
- Nhân viên cơ bản (dưới 2 năm): 12.000.000 - 18.000.000 VND/tháng
- Chuyên viên cấp cao (trên 5 năm): 30.000.000 - 50.000.000 VND/tháng
- Quản lý (Manager): 50.000.000 - 80.000.000 VND/tháng

Source: page 0 (chinh_sach_luong.txt), page 0 (bao_hiem_xa_hoi.txt)
Time - Retrieval: 0.07s | Generation: 115.75s | Total: 115.81s

```

Figure 4.2: Local Vietnamese HR chatbot statistics after rebuilding the index.

## 4.4 Limitations and Threats to Validity

This study has three main limitations:

- **Synthetic Data and Limited Verification:** The corpus is based on demonstration HR policies rather than an approved handbook from a real company. In addition, the generated Q&A pairs were not exhaustively verified against every source sentence.
- **Small-Scale and Heuristic Evaluation:** The retrieval and generation evaluations use small test sets, and the automated checks mainly measure topical relevance rather than factual correctness, entailment, or legal validity.
- **Prototype-Level Deployment Constraints:** The system has not been deployed in a real company environment. Practical concerns such as access control, audit logging, secure storage, Vietnamese tokenization quality, and hardware dependence remain outside the current prototype scope.



## 5. Conclusion and Perspective

This project implemented a private Vietnamese HR assistant in which the documents, retrieval index, and language model run locally. Hybrid BM25–vector retrieval improved top-1 source selection from 6/10 to 10/10 on a small internal test, showing that lexical and semantic rankings are complementary for short queries.

The H200 experiment generated 4,900 unique Q&A pairs and trained a Phi-3-Mini QLoRA adapter in 5.3 minutes. Fine-tuning improved the automated topicality and format score and reduced direct inference latency; however, manual inspection still found incorrect policy facts. Therefore, the main conclusion is not that fine-tuning solved factual accuracy. Instead, controlled data generation made the model more domain-focused, while retrieval remained necessary to provide authoritative and inspectable facts.

Future work should include: (1) linking each generated answer to an exact supporting span, (2) using source-grounded reference answers and human HR review, (3) comparing base and fine-tuned GGUF models within the same RAG pipeline, (4) adopting Vietnamese word segmentation for BM25, and (5) testing the system with an approved real-company handbook. These steps would help transform the current proof of concept into a more credible evaluation of practical HR policy assistance.

## A. Reproducibility Map

The full source code, generated data, and raw model outputs are stored in the project repository rather than reproduced as lengthy listings in this thesis. Table A.1 maps each reported result to its corresponding local artifact.

**Table A.1:** *Local evidence artifacts used in this report.*

| Artifact                       | Evidence provided                                                                  |
|--------------------------------|------------------------------------------------------------------------------------|
| data/generated_qa_h200.jsonl   | 4,900 accepted questions, answers, source topics, and batch metadata.              |
| scripts/validate_qa.py         | Reproducible keyword relevance, red-flag, length, and duplicate checks.            |
| finetune_run.log               | Model ID, hyperparameters, example count, duration, final loss, and export events. |
| models/phi3-mini-hr-finetuned/ | LoRA configuration, tokenizer assets, and adapter weights.                         |
| models/phi-3-mini.gguf         | 2.32 GB Q4 artifact with HR fine-tune metadata.                                    |
| data/eval_scores.csv           | Forty rows containing outputs, flags, scores, and latency.                         |
| data/eval_report.txt           | Base/fine-tuned comparison and answer excerpts.                                    |
| scripts/reindex\_chromadb.py   | Rebuilds the 20-policy Vietnamese ChromaDB index; default output is 258 chunks.    |
| src/hybrid\_retriever.py       | BM25, vector search, and RRF implementation.                                       |

The following commands reproduce the dataset validation and thesis build from the repository root:

```
python scripts/validate_qa.py --input data/generated_qa_h200.jsonl
cd report
compile.bat
pdffinfo thesis.pdf
```

Reproducing inference requires sufficient memory and the Python/Unsloth environment used by the evaluation script. The CSV and text report preserve the completed H200 outputs, so report verification does not require renting a new GPU instance.

## Bibliography

- [1] P. Lewis et al., “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,” *Advances in Neural Information Processing Systems*, 2020. <https://doi.org/10.48550/arXiv.2005.11401>
- [2] S. Robertson and H. Zaragoza, “The Probabilistic Relevance Framework: BM25 and Beyond,” *Foundations and Trends in Information Retrieval*, vol. 3, no. 4, pp. 333–389, 2009. <https://doi.org/10.1561/15000000019>
- [3] G. V. Cormack, C. L. A. Clarke, and S. Buettcher, “Reciprocal Rank Fusion Outperforms Condorcet and Individual Rank Learning Methods,” *SIGIR*, pp. 758–759, 2009. <https://doi.org/10.1145/1571941.1572114>
- [4] Microsoft, “Phi-3 Technical Report: A Highly Capable Language Model Locally on Your Phone,” 2024. <https://doi.org/10.48550/arXiv.2404.14219>
- [5] N. Reimers and I. Gurevych, “Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks,” *EMNLP*, pp. 3982–3992, 2019. <https://doi.org/10.18653/v1/D19-1410>
- [6] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, “QLoRA: Efficient Finetuning of Quantized LLMs,” *NeurIPS*, 2023. <https://doi.org/10.48550/arXiv.2305.14314>
- [7] E. J. Hu et al., “LoRA: Low-Rank Adaptation of Large Language Models,” *ICLR*, 2022. <https://doi.org/10.48550/arXiv.2106.09685>
- [8] Chroma Inc., “ChromaDB Documentation,” 2023. <https://docs.trychroma.com/>
- [9] National Assembly of Vietnam, “Labour Code, Law No. 45/2019/QH14,” 2019. <https://thuvienphapluat.vn/van-ban/Lao-dong-Tien-luong/Bo-Luat-lao-dong-2019-388600.aspx>
- [10] D. Q. Nguyen and A. T. Nguyen, “PhoBERT: Pre-trained Language Models for Vietnamese,” *Findings of EMNLP*, pp. 1037–1042, 2020. <https://doi.org/10.18653/v1/2020.findings-emnlp.92>
- [11] K. Nguyen, V. Nguyen, A. Nguyen, and N. Nguyen, “A Vietnamese Dataset for Evaluating Machine Reading Comprehension,” *COLING*, pp. 2595–2605, 2020. <https://doi.org/10.18653/v1/2020.coling-main.233>